Midsemester Progress Report

Team Gold

CS411W

03-20-2026

Weston

- Built the database layer from the ground up, designing the initial MySQL schema and iterating on it multiple times as the project's needs evolved
    - Introduced auto-incremented primary keys across tables
    - Added a dedicated sessions table for user session management
    - Dropped unnecessary tables to streamline the overall structure
- Integrated geospatial support into the database, enabling location-aware features across the app
    - Added a current_location column with spatial indexing
    - Configured the column to store coordinates in a format compatible with MySQL's spatial functions
- Collaborated closely with Carlos to align on backend architecture and ElysiaJS best practices
    - Met regularly to discuss structure, task delegation, and backend needs as requirements changed
    - Ensured the codebase was organized for maintainability as the team scaled up
- Built out the backend service layer in ElysiaJS, writing service files covering the core functionality of the app
    - Wrote service functions for user sessions, forum posts, and comments
    - Implemented logic to filter forum posts within 15 miles of the user
    - Tied post creation to active user session tokens for security

- Consistently led Monday team meetings and kept the group organized throughout the project
    - Broke down and delegated tasks to the group via Discord
    - Handled team documentation including weekly reports

Carlos

- Contributed to the backend API by helping design routing structure and modeling data to match the database schema
    - Worked alongside Weston to ensure data structures were consistent between the service layer and the database
    - Helped establish the conventions the backend routing follows throughout the project
- Implemented the core module files for the app's primary features
    - Built model.ts and index.ts for the users, authentication, forums, and notifications modules
    - Gave the backend a consistent and organized foundation for the rest of the team to build on
- Refactored the backend architecture to align with ElysiaJS best practices
    - Improved overall code readability and structure across multiple iterations
    - Ensured the backend would remain maintainable as new features were added
- Set up Swagger for in-browser endpoint testing
    - Gave the team a reliable way to verify backend routing and functionality
    - Reduced reliance on a fully connected frontend to test backend behavior
- Began bridging the frontend and backend together
    - Started with the user login and signup pages, handling the initial wiring between UI and API
    - Worked through bugs encountered during integration
- Laid early groundwork for user security and database testing
    - Researched authentication methods and password hashing to ensure secure user handling from the start
    - Created mock data to facilitate database testing before full integration was complete

Josh

- Laid the foundation for the entire frontend early in the project
  - Initialized Expo in the repository and set up the app's menu structure
  - Created placeholder screens for the team to build features on top of
- Got Mapbox integrated into the project and set up for the whole team
  - Created the original Mapbox account and shared API tokens with the team
  - Implemented the Mapbox SDK into the React Native environment
  - Documented and communicated every local environment change so teammates could stay unblocked
- Built the initial MapView component with core functionality the team could build on
  - Implemented satellite view as the default map style
  - Added zoom, pan, and automatic location centering controls
- Contributed to the user authentication flow on the frontend side
  - Developed the Login and Signup pages
  - Implemented basic email verification
- Most recently extended the map with search and community features
  - Implemented map search connected to the Mapbox query API
  - Added markers for search results on the map
  - Overlaid forum feed markers onto the map to visually connect community posts with their locations

Tony

- Set up user location tracking early in the project and kept the team unblocked
  - Configured location tracking through Mapbox
  - Shared environment variables with the group so everyone could run the feature locally

- Enhanced the core screens with production-ready UX polish
  - Added loading states and empty state messaging to the Forum, Map, and Profile screens
  - Made the app feel responsive and complete even before full backend data was flowing
- Built reusable UI components that are used across the app
  - Created a SearchBar component for location searching on the map
  - Built a MapLoadingOverlay to provide visual feedback while the map initializes
  - Updated icon mappings across the app to stay consistent as new screens were added
- Implemented clear route functionality on the map screen
  - Built the reset logic to clear the active destination and any set routes
  - Wired the clear route behavior into the onSelected handler
  - Added the corresponding UI control button to the map interface

Derrick

- Built the community forum interface from scratch
  - Created the core forum feed layout for browsing posts
  - Developed the post submission UI so users can contribute content to the platform
  - Integrated all forum components into the app's tab layout as a first-class navigation destination
- Ensured quality and compatibility throughout forum development
  - Tested forum UI functionality locally at each stage before merging
  - Added thorough comments and documentation to help the team understand the structure as backend integration begins
  - Coordinated with teammates to make sure the forum components matched the planned backend architecture
- Extended the forum with features that connect it to the rest of the app
  - Began wiring the forum to support navigation from map pins to related posts
  - Implemented the user account page, rounding out the core screens for a functional user experience

Nick

- Built the post detail screen and interactive comments UI
- Implemented global ForumContext with a shared addComment action, replacing local state across screens
- Refactored forum navigation and synced the forum feature branch with main